Event-Log Augmentation under User-Defined Process Constraints

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Abstract. Event logs produced by information systems are a key input for analyzing, monitoring, and improving process executions, including service-oriented ones. However, such logs are often scarce or incomplete due to technical, managerial, or legal constraints, which limits the reliability of downstream analytics. Event-log augmentation can help produce larger and richer logs, but augmented traces must remain meaningful by satisfying service- and process-level constraints (e.g., interaction protocols, compliance policies, or SLA-inspired rules). This paper introduces a constraint-aware technique for augmenting an initial event log with additional traces. The technique combines probabilistic models of observed process behavior with automaton-based representations of user-defined constraints. The goal is to generate traces that comply with the specified constraints while remaining behaviorally realistic and generalizing beyond the traces observed in the original log, thereby providing additional information for subsequent analyzes. We quantitatively assess the technique across five use cases, each assessed under increasingly stricter constraint sets derived from the original event log. The results show that our approach preserves compliance while increasing the diversity in the generated traces. We further evaluated one use case in the context of process discovery, comparing the fitness and precision of models mined from non-augmented logs, logs augmented without considering constraints, and logs augmented with our constraint-aware technique. The results indicate that considering constraints during augmentation leads to discovered models that better balance fitness and precision.

Keywords: Event-log augmentation · Generative methods · Process mining · User-defined constraints · Finite State Automata

1 Introduction

Process mining aims to understand and improve processes by analyzing their transactional data, which describes how individual executions unfold [1]. Such transactional data are typically captured in event logs, i.e., collections of traces that each represent one execution of the process. A trace is a sequence of events, where each event records the start or completion of a specific activity, together with its timestamp and other event-related perspectives.

Process mining is feasible only when an event log is available. As noted by Zimmermann et al. [16], data availability remains one of the major challenges facing both practitioners and researchers in the field. The problem is particularly severe for approaches based on machine or deep-learning models, which are inherently "data hungry". In process mining, this is often the case for simulation, Predictive Process Monitoring, and Prescriptive Analytics, e.g. [13]. These techniques rely on AI-based models that require substantial amounts of training data. Consequently, when the original event log is

small, these techniques cannot be applied effectively. This motivates the need for event-log augmentation methods that enrich the original set of traces with newly generated ones.

The problem of event-log augmentation has recently gained momentum (cf. Section 2). However, the proposed approaches are not supported in settings in which process analyst would like to impose constraints when generating new traces. Where possible, applying constraints offers several advantages, enabling the enforcement of potentially new domain rules and semantics. This also prevents the introduction of unrealistic behavior in the generated traces and ensures that they are meaningful. Then, it ultimately means that compliance is ensured in the augmented logs, with evident improvements in the usefulness of the generated traces for downstream tasks. For instance, if an event log is augmented without imposing constraints, when used in prescriptive process analytics, recommendations can be provided that violate the process' constraints.

This paper provides a technique to augment event logs under process' constraints that is based on finite-state automata (FSA) theory. In a nutshell, process constraints are given as a set of finite-state automata. Given an event log $\mathcal{L}$ to be augmented, a stochastic finite-state automaton is generated to represent the behavior observed in $\mathcal{L}$. This stochastic finite-state automaton is intersected with the constraint automata, thus obtaining a new stochastic finite-state automaton that is only capable of generating traces compliant with the constraints.

Experiments were conducted with five event logs and three progressively restrictive constraint sets mined using a technique from the state-of-the-art, and compared with three baselines: i) the first where the non-compliant traces are filtered out before generating a stochastic finite-state automaton that does not enforce the constraints, ii) a state-of-the-art technique of conditioned data augmentation for process mining [10], and iii) a constraint-driven technique that only generates an event-log on the basis of the mined constraints [3]. From a theoretical viewpoint, it is worth highlighting that the first baseline and the work in [10] do not always guarantee that the constraints are not violated (cf. discussion at the end of Section 5.3). From the practical viewpoint, our proposal guarantees a higher level of generalization if compared with those baselines, while the similarity to the original event logs is largely retained at the same level. Generalization is here meant to indicate variability, which can be computed through the entropy-based metrics proposed by Back et al. [4].Similarity is measured through the 2-gram log distance metrics [5], which measures the average minimal edits required to align traces between logs.

The remainder of this paper is organized as follows. Section 2 reviews the related literature on event-log augmentation.Section 3 introduces the foundational definitions and core concepts relevant to event-log augmentation. Section 4 details the proposed constraint-aware augmentation framework. Section 5 discusses the experimental pipeline, discussing the results in Section 5.3. Section 7 concludes the paper and outlines limitations and directions for future research.

2 Related Works

Traditional data augmentation techniques, commonly used in domains like computer vision and natural language processing, often assume stochastic independence between samples [14]. However, such assumptions do not hold for event logs, which are in-

herently arranged in sequences, where temporal and contextual dependencies are crucial [1].

Käppel et al. have proposed a first approach to event log augmentation in [11], where the event log augmentation framework relies on random noise-based transformations, while van Straten et al. [15] introduced an augmentation framework for event logs inspired by techniques from natural language processing, based on replacement of process activities.

In parallel, a growing body of work has emerged around Business Process Simulation (BPS) [6]. However, while BPS aims to replicate real process behavior for what-if analysis, event-log augmentation pursues a complementary objective: generating synthetic event logs that enrich observed process data, providing a broader empirical foundation for process analysis and data-greedy methods. Despite progress in both areas, no existing framework enables users to selectively specify process constraints, and to generate event logs from them while maintaining coherence with the original event log.

The generation of constraint-compliant traces requires a method to enforce process constraints. To this extent, the DECLARE constraint-based process modeling language [12] provides a viable option. Di Ciccio et al. [3] introduced efficient discovery algorithms for DECLARE constraints from event logs, also capable of generating traces from a mined DECLARE model. However, these constraints cannot be manually set by process analysts and the generation only depends on the set of mined constraints. More recently, a data-aware extension of declarative process mining that generates both the control-flow and data perspectives, has been proposed in [9], enabling more controlled and reproducible experimental setups. Nevertheless, the underlying generation mechanism remains solely driven by mined declarative constraints rather than by the statistical or structural properties of the event log itself.

Graziosi et al. [10] propose a method for event-log generation that is based on Conditional Variational Autoencoders (CVAE). They rely on flagging the original traces as "regular" or "deviant", guiding the generation of "regular" traces, only. However, the set of constraints is not directly provided as input when the CVAE model is trained: the model is expected to learn these constraints from a differential analysis between "regular" or "deviant". Therefore, there is no guaranty that the model learns the exact constraints: Thus, it cannot be excluded that traces violating these constraints are generated. In fact, the experimental results reported in Section 5.3 show that a large share of the generated traces violate the constraints, especially when several constraints are modelled.

In addition to its competitive performance and efficiency, the introduced framework is inherently white-box, making the integration of process constraints considerably simpler compared to black-box deep-learning approaches. These considerations motivate our choice of the Stochastic FSA–based method as the foundation upon which to develop our constraint-aware generator.

3 Preliminaries

The starting point for a process mining-based system is an *event log*. An event log is a multiset of *traces*. Since this paper only focuses on the activity labels when traces are generated, each trace is abstracted as a sequence of activities, each describing a particular execution of a *process instance*:

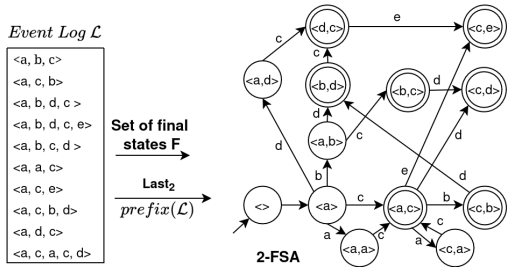

Fig. 1: Example of generation of a 2-FSA leveraging an event log $\mathcal{L}$ and a $last_2$ function.

Definition 1 (Traces & Event Logs). *Let $\mathcal{A}$ be the set of possible activities, and let $\mathcal{B}(\mathcal{A}^*)$ be the set of all multisets of sequences in $\mathcal{A}$. A trace $\sigma \in \mathcal{A}^*$ is defined as a finite sequence of activities. An event log $\mathcal{L} \in \mathcal{B}(\mathcal{A}^*)$ is a multiset of sequences of activities in $\mathcal{A}$.*

To capture the different states in which a non-completed trace can be, we use the *prefixes*, i.e., a sequence of events that represent the process execution up to a certain point. Given a trace $\sigma = \langle a_1, \ldots, a_n \rangle$, the set of possible prefixes is defined as: $prefix(\sigma) = \{\langle\rangle, \langle a_1 \rangle, \langle a_1, a_2 \rangle, \ldots, \langle a_1, \ldots, a_n \rangle\}$. Analogously, we are also interested in annotating the set of prefixes of the traces in a given event log. The multiset containing all the prefixes of all the traces in an event log will be referred to as $prefix(\mathcal{L})$, i.e. $prefix(\mathcal{L}) = \uplus_{\sigma \in \mathcal{L}} prefix(\sigma)$.[1]

The remainder of this section illustrates the formal foundation of the event-log augmentation framework based on a stochastic finite-state automaton. The formalization relies on a helper function $last_k$ with $k \in \mathbb{N}$. Given a trace $\sigma = \langle a_1, \ldots, a_n \rangle$, $last_k(\sigma)$ returns the sub-trace consisting of the last k events $\langle a_{n-k}, \ldots, a_n \rangle$; if $n < k$, the same trace σ is returned.

We can now formally define the concept of k-Finite-state Automaton:

Definition 2 (k-Finite-state Automaton (k-FSA)). *Let $\mathcal{L}$ be an event log defined over a set $\mathcal{A}$ of activities. Let k be a natural number. The Stochastic k-Finite Automaton is defined as $(S, \mathcal{A}, \delta, \langle\rangle, F)$, which denotes a finite state automaton, consisting of:*

- *a state set $S = \bigcup_{\sigma \in prefix(\mathcal{L})} last_k(\sigma)$,*
- *a transition set $\mathcal{A}$,*
- *a transition function $\delta : S \times \mathcal{A} \not\to S$ s.t.*
  - $\forall a \in \mathcal{A},\ \langle a \rangle \in prefix(\mathcal{L}) \Rightarrow \delta(\langle\rangle, a) = \langle a \rangle$
  - $\forall \langle a_1, \ldots, a_n, a_{n+1} \rangle \in prefix(\mathcal{L})$.,
    $\delta(last_k(\langle a_1, \ldots, a_n \rangle), a_{n+1}) = last_k(\langle a_1, \ldots, a_{n+1} \rangle)$
- *the empty trace $\langle\rangle$ is the initial state,*
- *the set $F = \bigcup_{\sigma \in \mathcal{L}} last_k(\sigma)$ of final states.*

The following example illustrates how a 2-FSA is created starting from an event log:

[1] Symbol $\uplus$ indicates the union of multisets.

Example 1 (2-FSA). An example of 2-FSA is shown in Figure 1. Starting from an event log $\mathcal{L}$ composed of 10 complete traces, the multiset of prefixes $prefix(\mathcal{L})$ and the function $last_k$ are leveraged to infer the set of states $S = \{\langle\rangle, \langle a\rangle, \langle a, b\rangle, \langle a, a\rangle, \langle a, c\rangle, \langle a, d\rangle, \langle b, c\rangle, \langle b, d\rangle, \langle c, b\rangle, \langle c, e\rangle, \langle d, c\rangle, \langle c, d\rangle, \langle c, e\rangle\}$. The set of final states $F \subset S$ is then created from $\mathcal{L}$ and reported in figure with a double circle.

The transitions of a k-FSA can be annotated with the probability of occurrence, thus producing a Stochastic k-FSA:[2]

Definition 3 (Stochastic k-Finite-state Automaton (Stochastic k-FSA)). *Let $\mathcal{L}$ be an event log defined over a set $\mathcal{A}$ of activities. Let k be a natural number. A Stochastic k-Finite-state Automaton is a tuple $(S, \mathcal{A}, \delta, \langle\rangle, F, P)$, where $(S, \mathcal{A}, \delta, \langle\rangle, F)$ is a k-FSA and $P : S \times \mathcal{A} \not\to [0, 1]$ is a function that for each $(s, a) \in \mathrm{dom}(\delta)$, returns the probability $P(s, a)$ of firing transition (s, a).*

Note that for any final states $s \in F$, $0 \leq \sum_{a \in \mathcal{A}} P(s, a) \leq 1$. This occurs because the trace generation stops when reaching s with probability $1 - \sum_{a \in \mathcal{A}} P(s, a)$ and goes on with probability $\sum_{a \in \mathcal{A}} P(s, a)$.

Leveraging the defined probability function P, the Stochastic k-FSA can be traversed to stochastically generate sequences of symbols that conform to the behavior captured by the underlying finite-state automaton. Consequently, each generated sequence is guaranteed to be valid with respect to the automaton structure, while the stochastic nature of the transitions enables the realistic modeling and simulation of process variability observed. The probabilities are derived directly from the original event log, ensuring that the resulting model accurately reflects the empirical control-flow dynamics observed in the real data.

Given an event log $\mathcal{L}$ and its corresponding k-FSA $(S, \mathcal{A}, \delta, \langle\rangle, F)$, the probability function of the corresponding Stochastic k-FSA $(S, \mathcal{A}, \delta, \langle\rangle, F, P)$ is defined as follows. For each $(s, a) \in \mathrm{dom}(\delta)$, the probability $P(s, a)$ is the ratio between *(i)* the number of prefixes that ends with sequence s followed by a and *(ii)* the number of prefixes in which the trace suffix is s:[3]

$$P(s, a) \coloneqq \frac{\left|\{\sigma' \in \mathrm{prefix}(\mathcal{L}) : \mathrm{last}_{k+1}(\sigma') = s \oplus \langle a\rangle\}\right|}{\left|\{\sigma' \in \mathrm{prefix}(\mathcal{L}) : \mathrm{last}_k(\sigma') = s\}\right|} \tag{1}$$

The probabilistic transitions in the automaton are directly grounded in the observed frequency of activity successions in the event log. This data-driven probability assignment allows the automaton to generate new traces that not only comply with the structural rules encoded in the k-FSA but also preserve the stochastic properties of the original process behavior.

Let $\mathcal{Z} = (S, \mathcal{A}, \delta, q_0, F, P)$ be a Stochastic k-FSA. A trace $\sigma = \langle a_1, a_2, \cdots, a_n\rangle \in \mathcal{A}^*$ is admissible by $\mathcal{Z}$, and thus can be generated, iff there a state sequence $s_0 \xrightarrow{a_1} s_1 \xrightarrow{a_2} s_2 \cdots \xrightarrow{a_n} s_n$ such that $s_i = \delta(s_{i-1}, a_i) \in S$ and $s_n \in F$ for all $i \in \{1, \ldots, n\}$. If σ is admissible, the probability of generating σ is $\prod_{i=1}^{n} P(s_{i-1}, a_i)$.

Example 2 (Stochastic 2-FSA). Figure 2 shows the Stochastic 2-FSA obtained from the 2-FSA in Figure 1. Using the multiset $prefix(\mathcal{L})$, transition probabilities are calculated

[2] Given a function f, in the remainder, $\mathrm{dom}(f)$ denotes the domain of f.

[3] Symbol $|\ |$ is here overridden to be applied to multisets, namely to account for the sequence frequencies, and $\oplus$ is used to denote the concatenation of sequences.

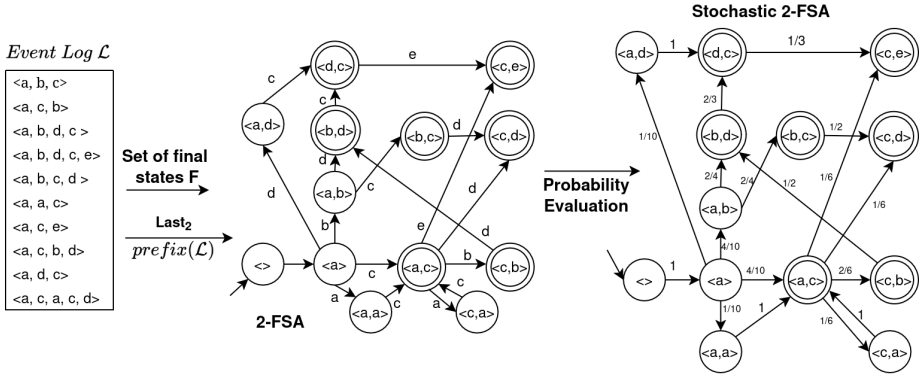

Fig. 2: Example of generation of a Stochastic 2-FSA, starting from the same event log proposed in Figure 1.

according to the formula in Eq. 1. For instance, at the state $\langle a \rangle$, the probability of firing the transition labeled c is $\frac{4}{10}$. This is because the next state $\langle a, c \rangle$ appears 4 times right after $\langle a \rangle$, in the traces $\langle a, c, b \rangle$, $\langle a, c, e \rangle$, $\langle a, c, b, d \rangle$, and $\langle a, c, a, c, d \rangle$ out of 10 occurrences of the state $\langle a \rangle$, since every trace starts with $\langle a \rangle$. In contrast, consider the state $\langle d, c \rangle$, which is followed only once by $\langle c, e \rangle$ in the log. Since $\langle d, c \rangle$ occurs 3 times in total, and 2 of these are final states, the probability of transitioning with e is $\frac{1}{3}$, while the probability of terminating generation at this state is $\frac{2}{3}$. The framework also supports looping behavior, as illustrated by state $\langle a, c \rangle$, which admits a transition labeled a with probability $\frac{1}{6}$, leading to state $\langle c, a \rangle$ from which a return to $\langle a, c \rangle$ is guaranteed. While such loops are structurally permitted, their occurrence remains statistically unlikely given the low associated transition probability.

4 The Framework Proposal

This section introduces the core framework for event-log augmentation under user-defined process constraints. These constraints are expected to be defined by process analysts in the form of finite-state automata:

Definition 4 (Constraint Automaton). *Let $\mathcal{A}$ be a finite set of activities. A Constraint Automaton over a set $\mathcal{A}$ of activities is a tuple $(Q, \mathcal{A}, \delta, q_0, F)$ where:*

- *Q is a finite set of states,*
- *$\delta : Q \times \mathcal{A} \nrightarrow Q$ is the transition function,*
- *$q_0 \in Q$ is the initial state,*
- *$F \subseteq Q$ is the set of final states.*

Note that the above definition of constraint automata is compliant with every constraint definition supported by the DECLARE language [2], although it is more general. Users can thus provide the constraints through the graphical notation of DECLARE. The following shows examples of constraint automata:

Example 3 (Constraint Automata). Figure 3 depicts two constraint automata, corresponding to two constraints: C_1 = {if d occurs, e has to eventually occur after d},

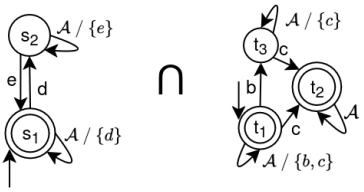

Fig. 3: Example of constraint automata: $C_1 = \{$if d occurs, e has to eventually occur after $d\}$, and $C_2 = \{$if b occurs, c has to occur before or after $b\}$

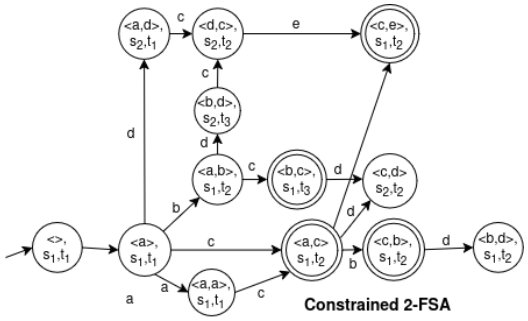

Fig. 4: Example of a Constrained 2-FSA.

and $C_2 = \{$if b occurs, c has to occur before or after $b\}$. The automaton for C_1 consists of two states, s_1 and s_2, where only s_1 is accepting. In contrast, the automaton for C_2 has three states: t_1 represents when neither b nor c has occurred; t_2 represents the situation where c has occurred (removing restrictions on b); and t_3 denotes that b has occurred but c has not yet appeared. The two automata will be then intersected to form a Constraint Automaton whose states are pairs drawn from the original automata's states. A state in this automaton is accepting if and only if both constituent states are accepting.

Definition 5 (Constrained k-FSA). *Let $\mathcal{A}$ be a set of activities, and let $\mathcal{C} = \{(Q_1, \mathcal{A}, \delta_1, q_{0,1}, F_1), \ldots, (Q_n, \mathcal{A}, \delta_n, q_{0,n}, F_n)\}$ be a set of constraint automata. The Constrained k-FSA is obtained by intersecting the Constraint Automata in $\mathcal{C}$ and a given k-FSA $(S, \mathcal{A}, \overline{\delta}, \langle\rangle, \overline{F})$ and consists in the automaton $(Q, \mathcal{A}, \delta, q_0, F)$ defined as follows:*

- *$Q = S \times Q_1 \times Q_2 \times \cdots \times Q_n$ is the product state space,*
- *$q_0 = (\langle\rangle, q_{0,1}, q_{0,2}, \ldots, q_{0,n})$ is the initial state,*
- *the transition function $\delta : Q \times \mathcal{A} \nrightarrow Q$ is defined as follows:*
  For each state $(s, q_1, \ldots, q_n) \in Q$ and activity $a \in \mathcal{A}$, if $(s, a) \in \mathrm{dom}(\overline{\delta})$ and $(q_1, a) \in \mathrm{dom}(\delta_1)$ and $\ldots$ and $(q_n, a) \in \mathrm{dom}(\delta_n)$

$$\delta\big((s, q_1, \ldots, q_n),\, a\big) = \big(\overline{\delta}(s, a), \delta_1(q_1, a),\ \delta_2(q_2, a),\ \ldots,\ \delta_n(q_n, a)\big),$$

- *the set of final states is $F = \overline{F} \times F_1 \times F_2 \times \cdots \times F_n$.*

Example 4 (Constrained 2-FSA). The Constraint Automaton log, obtained by intersecting the automata C_1 and C_2 from the Example in 3, is then intersected with the 2-FSA from the Example 1. Figure 4 shows the resulting Constrained 2-FSA where states are within the Cartesian product of the states of the 2-FSA in Figure 1 and of C_1 and C_2. A tuple is a final state if and only if every component state from the input automata is final. In this case, the four identified final states are $(\langle c, e\rangle, s_1, t_2)$, $(\langle b, c\rangle, s_1, t_3)$, $(\langle a, c\rangle, s_1, t_2)$, and $(\langle c, b\rangle, s_1, t_2)$. Conversely, the states $(\langle c, d\rangle, s_2, t_2)$ and $(\langle b, d\rangle, s_1, t_2)$ have no possible outgoing transitions and are not final; such states are referred to as *dead ends*.

The use of a constrained k-FSA to augment an event log requires one to associate transitions with probabilities, yielding a Stochastic Constrained k-FSA:

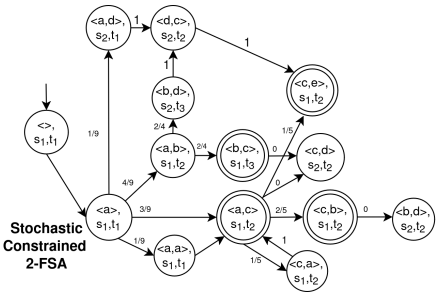

Fig. 5: Visual representation of the Stochastic Constrained 2-FSA, obtained from the Constrained-2FSA in Figure 4 and the Stochastic 2-FSA in Figure 2.

Definition 6 (Stochastic Constrained k-FSA). *Let $\mathcal{A}$ be a set of activities, and let $\mathcal{C}$ be a set of constraint automata. Let $\mathcal{K}$ be a k-FSA, and let $\mathcal{S} = (\mathcal{K}, P)$ be an accordant Stochastic k-FSA. The Stochastic Constrained k-FSA of $\mathcal{C}$ and $\mathcal{S}$ is a tuple $(Q, \mathcal{A}, \delta, q_0, F, P)$ where $(Q, \mathcal{A}, \delta, q_0, F)$ is the Constrained k-FSA of $\mathcal{C}$ and $\mathcal{K}$ and $P : Q \times \mathcal{A} \not\to [0, 1]$ is a function that assigns a probability $P(q, a)$ to each pair (q, a) in the domain of δ.*

The function P_c of a Stochastic Constrained Automata $(Q_c, \mathcal{A}, \delta_c, q_{0_c}, F_c, P_c)$ of a set $\mathcal{C}$ of constraint automata and a Stochastic k-FSA $(S, \mathcal{A}, \overline{\delta}, \langle\rangle, F_c, P)$ is computed as follows:

1. Assignment of a zero probability to transitions reaching a dead state. As also illustrated in Example 4, a Constrained k-FSA can contain some *dead* states q from which no final state is reachable, namely there is no sequence $\langle b_1, \ldots, b_n\rangle \in \mathcal{A}^*$ such that $\delta(\ldots\delta(\delta(q, b_1), b_2)\ldots, b_n) \in F_c$. First, we identify the set $Q_d \subset (Q_c \setminus F_c)$ of *dead* states. For each transition $(q, a) \in \mathrm{dom}(\delta_c)$ such that $\delta_c(q, a) \in Q_d$, $P_c(q, a) \coloneqq 0$.

2. Assignment of the probabilities to non-dead states. At step 1, the probabilities are set to zero for all transitions that lead to a dead state. To keep maintain consistency, the probabilities of the other transitions need to be scaled up: recalling that Q_d identifies the dead states, for each transition $((s, \overline{q}), a) \in \mathrm{dom}(\delta_c)$ such that $\delta_c((s, \overline{q}), a) \notin Q_c$, its probability is set as follows:

$$P_c((s, \overline{q}), a) \coloneqq P(s, a) \cdot \frac{\sum_{a' \in \mathcal{A} \,:\, (s,a') \in \mathrm{dom}(\overline{\delta})} P(s, a')}{\sum_{a' \in \mathcal{A} \,:\, (s,a') \in \mathrm{dom}(\overline{\delta}),\ \delta_c((s,q),a') \notin Q_c} P(s, a')} \tag{2}$$

Note how the ratio present in Equation 2 is always larger than or equal to one. If the probability of any transition $((s, q), a')$ is set to zero, the ratio is larger than one, thus distributing those probabilities to every transition that enables reaching a final state, proportionally to the original probabilities.

Example 5 (Stochastic Constrained 2-FSA). In this example, also depicted in Figure 5, the Stochastic 2-FSA created in example 2 and the Constrained 2-FSA created in example 4 are used to create the Stochastic Constrained 2-FSA. As reported in Definition 6, the construction of it implies two steps: in the first step the transitions reaching a dead state are assigned probability 0. This applied to the dead states $(\langle c, d\rangle, s_2, t_2)$

and $(\langle b, d\rangle, s_2, t_2)$ that have been assigned probability 0. Then, probabilities are assigned and rescaled since the transition from the state $(\langle a, c\rangle, s_1, t_2)$ to $(\langle c, d\rangle, s_2, t_2)$ had probability $\frac{1}{6}$, now the probabilities of the other transition that can be fired after $(\langle a, c\rangle, s_1, t_2)$ are multiplied by a factor of $\frac{6}{5}$.

5 Quantitative Evaluation

This section details the experiments that have been carried out to quantitatively evaluate the quality of the generated event logs in terms of adherence to and generalization of the original event log.[4]

5.1 Use Cases and Definition of Constraint Sets

The validity of the framework was assessed using five different processes, and the corresponding event logs were available:

Purchasing. A synthetic process related to a purchase-to-pay process, provided as part of the Fluxicon Disco tool related to a purchase-to-pay system.[5]

Production. A process dealing with a manufacturing production company (MP). It has been exported from an Enterprise Resource Planning (ERP) system [6].

Consulta. The Academic Credentials Recognition (ACR) process of a Colombian University.[7]

Lending. The synthetic process from which a generated log pertaining to a loan application process has been extracted.[8]

Emergency Dept. This real process has been provided by an Italian hospital emergency department, and it is private.

This section reports controlled experiments that aim to be objectively designed and run; therefore, the constraints are discovered from the event logs. In contrast, Section 6 reports a real-life use case, where the stakeholders provide constraints, as expected in a production environment. The constraints are discovered through the Declare miner implemented in `pm4py`.[9] It is well known that each Declare constraint can be transformed into a corresponding finite state automaton as per Definition 4 that accepts exactly those traces that satisfy the constraint [7]. In particular, we mined the constraints with a support larger than 0.8 and Conditional-Probability Increment Ratio (CPIR) larger than 0.85. The choice is in line with the guidelines in [12]. For each use case, we then chose the top 5, 25 and 50 constraints in terms of support, aiming to illustrate whether our framework scales with larger and larger sets of constraints and whether the benefits were more or less evident with a larger set of constraints.

[4] Implementation available at https://anonymous.4open.science/r/Constrained_data_aug-6B86/README.md

[5] https://fluxicon.com/academic/material

[6] https://doi.org/10.4121/uuid:68726926-5ac5-4fab-b873-ee76ea412399

[7] https://zenodo.org/records/5734443

[8] https://zenodo.org/records/8059489

[9] The `pm4py`'s implementation of the the Declare miner is discussed at https://processintelligence.solutions/pm4py/api/api/pm4py.algo.discovery.declare.algorithm.html

5.2 Experimental Settings

A set of experiments were run on the combinations of the use cases and sets of defined constraints that were mentioned above.

The ultimate goal is to illustrate that our framework based on Constrained Stochastic k-FSA is able to increase the level of generalization while maintaining a similar level of distance between the generated and the original event log. Typically the distance with the original event log and the generalization are opposite dimensions, but we aim to show that our framework enables to obtain a better trade-off, if compared with what state-of-the-art techniques are able to achieve.

Experimental Metrics. The level of generalization was measured through the trace entropy computed in the event logs generated [4]. Trace entropy measures the diversity of traces in an event log. It is computed by considering the relative frequency of each distinct trace and applying the standard entropy formula for sequences. A low trace entropy indicates that the log contains mainly similar repeated traces, while a high trace entropy indicates greater variability among the traces [4]. The distance to the original event log was computed through the 2-gram log distance (2-gram), introduced in [5], between the generated event log and $\mathcal{L}_{test}^{\mathcal{C}}$, previously introduced as the sub-log of the traces of $\mathcal{L}_{test}$ that comply with the set $\mathcal{C}$ of constraints of interest in the experimental run. The 2-gram log distance compares two event logs by counting how often each pair of consecutive activities occurs on the traces of both logs. Then it computes the normalized sum of the absolute differences between these 2-gram frequencies. Alternatively, we could have adopted the Control-Flow Log Distance (CFLD) measure [5]. However, CFLD is computationally demanding in terms of both time and memory demands, and it could not be computed for several of the use cases considered in our experiments. For this reason, we relied on the 2-gram distance, which *"effectively captures the effect of perturbations along the control-flow perspective, achieving similar results as the Control-Flow Log Distance (CFLD) measure [...] while being much more computationally efficient"* [5].

Experimental Methodology. The event log $\mathcal{L}$ of each use case was divided into a training log $\mathcal{L}_{train}$ and a test log $\mathcal{L}_{test}$ following a temporal splitting. The training log $\mathcal{L}_{train}$ was constructed from the first 50% of the traces. For each use case, log $\mathcal{L}_{train}$ was used with each set $\mathcal{C}$ of the mined constraint (cf. previous paragraphs) to build our stochastic Constrained k-FSA where k was set to vary between 1 and 10. The determination of the best k was made as follows. For each set $\mathcal{C}$ of constraints, we retained the sub-log $\mathcal{L}_{test}^{C}$ of the trace in $\mathcal{L}_{test}$ that did not violate $\mathcal{C}$. For each $k \in [1, 10]$, we then generated 10 synthetic event logs with the same number of traces as $\mathcal{L}_{test}^{C}$ and computed the average trace entropy of these 10 synthetic event logs and the average 2-gram log distance between these 10 synthetic event logs and $\mathcal{L}_{test}^{C}$. For each use case and set $\mathcal{C}$, we ultimately retained the Stochastic Constrained k-FSA with the best balance between entropy and log distance.

Comparison Baselines. The synthetic event logs generated through our Stochastic Constrained k-FSA were compared with those obtained through three baselines from the state of the art:

1. A non-stochastic constrained k-FSA with same value for k as used for our Stochastic Constrained k-FSA. In this setting, the traces that violate the constraints are filtered out before building the non-stochastic k-FSA.

use case	5 constraints		25 constraints		50 constraints	
	S k-FSA	CVAE	S k-FSA	CVAE	S k-FSA	CVAE
Consulta	0%	12.7%	6.2%	17.2%	9.0%	76.4%
Production	0.5%	0%	8.3%	85.4%	9.3%	90.6%
Purchasing	0%	30.6%	26.4%	46.5%	27.0%	50.8%
Emergency Dept	0%	8.2%	9.2%	5.9%	10.2%	8.0%
Lending	0%	5.2%	0%	9.7%	0%	67.3%

Table 1: Percentage of generated traces non-conformant to constraints by use case, scenario, and constraint setting. We only report for the Stochastic k-FSA with k finite and optimized, and for CVAE becase they are the only baselines that do not guarantee compliance in the generated traces. The reported values are the mean of 10 repetitions.

2. The CVAE method by Graziosi et al. [10] flagging the non-conformant traces as "deviant" and the conformant traces as "regular".
3. The RuM toolkit by Alman et al. [3], where the Stochastic k-FSA is replaced by Declare model, which is mined from the event log. Namely, the constraints of the mined Declare model are intersected with the constraints provided by the user.

In addition to the above, we also considered our Stochastic Constrained k-FSA and a non-stochastic constrained k-FSA by setting $k = \infty$ as additional baselines. The expectation is that $k = \infty$ corresponds to the setting in which traces are simply sampled from the training logs, thereby resulting in the lowest levels of generalization.

5.3 Experimental Results

The first assessment focused on the baseline based on Stochastic k-FSA (S k-FSA) and CVAE, which do not provide a formal guaranty that the generated traces comply with the set of constraints. This is illustrated in Table 1, where, for each use case, the percentage of non-compliant traces with respect to the respective set of constraint is shown. The experiment on each use case and set of constraints was repeated 10 times to generate so many event logs. As expected, these two baselines can generate non-compliant traces, and this grows as the constraints are more and more restrictive. Experiments are shown that CVAE tends more than S k-FSA to generate non-compliant traces.

The numerical results are visualized in two graphs in Figure 6, where results are separately reported for each use case but they are aggregated by the different sets of constraints employed (namely the top 5, 25 and 50 constraints discovered through declare miner). These two comparative views simplify the analysis, compared with the detailed reporting of the raw values for each use case and set of constraints. However, the latter is available in the appendix that is part of the extended version available in the repository at link .

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Figure 6 reports the difference in the entropy or 2-gram values of each baseline, compared to our method based on stochastic constraint k -FSA, normalized with respect to the value obtained by our method. Namely, if x and y are the values obtained respectively by a baseline method and our method based on k-FSA, this normalized difference is therefore reported as $(x - y)/x$. For each use case and comparison baseline, the difference in entropy and 2-gram was computed separately for the different sets of constraints, then summarized by its median value, with the whiskers marking the minimum and maximum across those three sets of constraints. As mentioned in Section 5.2, k was optimized and changed for each use case: k was set to 3 for the use cases *Consulta*, *Purchasing* and *Emergency Department* and $k = 2$ for *Production*, *Cvs*

Use Case	SCk-FSA	SC∞-FSA	baseline k	S ∞-FSA	CVAE	RuM
Consulta	7min	7min	2min	2min	15min	3min
Production	5min	5min	1min	1min	70min	2min
Purchasing	6min	6min	1min	1min	25min	2min
Cvs	19min	19min	8min	8min	16h	4min
Lending	22min	22min	11min	11min	1d 4h	9min
Emergency Dept	54min	54min	29min	29min	1d 9h	15min

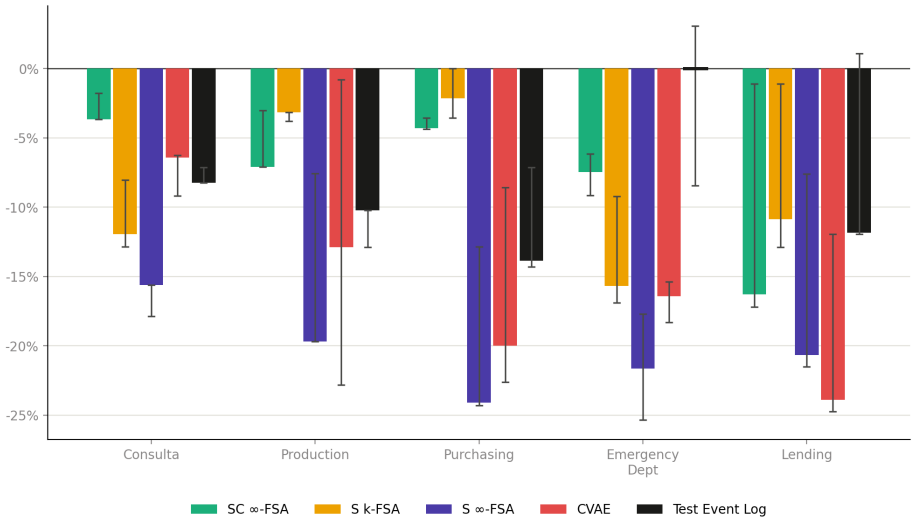
Table 2: Computational time required for training the approach and to simulate 10 different event logs.

and *Lending*. RuM was excluded from both graphs since its values were consistently far worse outside the range of the other methods, compressing the differences with the other baselines into an unreadable scale.

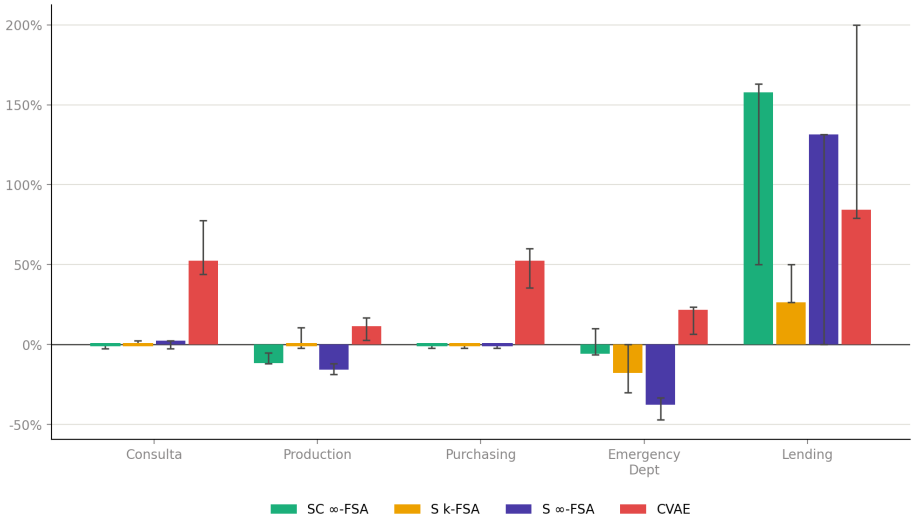
Figure 6a reports on the normalized differences in entropy between the baselines and our method based on k-FSA, per use case: negative values indicate a reduction in entropy of the baselines. Note that every method was used to generate the same number of traces as the test set for each configuration, ensuring comparability of the results. The differences are negative for every use case, meaning that the entropy of the event logs generated through our method based on Stochastic Constrained k-FSA tends to be higher, thus confirming our hypothesis that our method is better at generalizing when generating traces. As a further baseline for comparison, the normalized differences are also reported with respect to the entropy of the test log itself (see black bars): the negative difference with respect to the test event log, which contains the same number of traces, illustrates that our method is capable of generalizing more than the original event log.

Figure 6b conversely focuses on the normalized difference of the 2-gram metrics between the baselines and our method based on Constrained k-FSA: a negative value indicates that the value of the 2-gram metrics of the corresponding baseline is lower than what the generation through our method achieves, and lower value indicates more adherence of the generated traces to those of the original log. The graph shows that CVAE is well above our method in every use case, while the other baselines generally track close, except in *Lending*, where all baselines give higher 2-gram values, and *Emergency Department*, where the baselines based on Stochastic non-Constrained FSA (S k-FSA and (S ∞-FSA) show lower 2-gram values.The *Emergency Department* use case warrants a separate discussion: its median 2-gram value is 17% lower (i.e., better) than that of our method, whereas its entropy is 15.7% lower (i.e., worse) than our method. Hence, our approach does not appear to outperform the baseline S k-FSA. A plausible explanation is that the underlying process shows limited variability, as indicated by the fact that the entropy of the test event log is essentially equal to that of the event log produced by our method, which also yields the best entropy among the baselines. Under these circumstances, adding constraints does not substantially affect the generation process, but it does decrease the similarity to the test event log. However, recall that the baseline based on Stochastic non-constrained k-FSA may generate traces that do not comply the sets of constraints: in fact, the percentage of generated, non-compliant traces for this baseline is between 5.9% and 10.2% in every experiment with this use case.

We conclude the assessment with an average of the computation time of our method based on SC k-FSA and the baselines. Table 2 reports our findings for each case study. CVAE has very high computational times, while the others are characterized by times of the same order of magnitude. Notably, of the finite-state automata underlying the



(a) entropy



(b) 2-gram

Fig. 6: Entropy and 2-gram

Stochastic Constrained k-FSA does not introduce any significant computational overhead, even when one needs to intersect 50 FSAs (namely, one per constraint).

6 Application to a Real-life use case

We assessed the practical usefulness of the proposed method in a process-discovery scenario characterized by limited data availability. For each use case, we randomly selected 50 traces from the original event log, while the remaining traces were retained as

Use case	Constraint	Real	S-kFSA	SC-kFSA	SC–Real	SC–S
Production	`precedence(Final Inspection Q.C., Round Grinding - Machine 2)`	0.651	0.675	**0.737**	+13.2%	+9.2%
Consulta	`co_existence(Submit application for accreditation, Accreditation by course group)`	0.849	0.736	**0.907**	+6.8%	+23.2%
Lending	`precedence(Make Visit to Assess Collateral, Loan Denied)`	0.924	0.940	**0.962**	+4.1%	+2.3%

Table 3: Test-set F-scores obtained from the original 50-trace log and from logs augmented to 1,000 traces. The highest score for each use case is shown in bold.

an independent test set. We first applied the Inductive Miner, which is often used to discover process models that guaranty soundness, to this 50-trace log [1]. Its parameter α was optimized in the interval $[0, 1]$, with increments of 0.1, by maximizing the F-score, calculated as the harmonic mean of fitness and precision in the test event log [1]. The initial log was then increased to 1,000 traces using the proposed Stochastic Constrained k-FSA (SC-kFSA) or the corresponding unconstrained Stochastic k-FSA (S-kFSA). For each augmented log, a new process model was discovered using the noise threshold selected for the original log and evaluated on the held-out real traces.

As shown in Table 3, SC-kFSA achieved a higher best test-set F-score than both the original 50-trace log and S-kFSA in all three use cases. Because the best score was selected independently for each method, these maxima do not necessarily correspond to the same value of α. Compared to the original 50-trace log, the discovery via the event log with 1000 traces augmented through SC-k FSA produced methods with relative improvements of the F-score between 4.1% and 13.2%. Compared to the event log with 1000 augmented traces via the S-k FSA, the improved ranged between 2.3% and 23.2%. These results indicate that the observed benefit cannot be attributed solely to increasing the number of traces: explicitly enforcing the supplied constraints directs augmentation toward behavior that is more useful for discovering models that balance fitness and precision.

7 Conclusions

This paper presents a novel constraint-aware framework for event-log augmentation, integrating user-defined process constraints, expressed as finite-state automata, directly into the generative process that is based on them. The approach ensures strict compliance of all generated traces with the specified constraints while preserving trace variability, which is essential for effective process mining tasks such as simulation and predictive monitoring. Empirical evaluation across five use cases demonstrates a superior compliance-variability trade-off compared to baselines that filter non-compliant traces, generate event-logs directly from constraints and to a state-of-the-art frameworks, as validated by entropy-based metrics and 2-gram log distance.

The technique proposed here is naturally inapplicable if mutually conflicting constraints are included, yielding Stochastic Constrained FSA that accept no traces. However, techniques exist that can repair these sets of constraints, e.g. by Di Ciccio et al. [8].

The evaluation may be affected by a potential threat to validity: although the framework targets user-specified constraints, the quantitative controlled experiments obtain their constraint sets from the very event logs used to build the behavioral model. This

preserves objectivity but can introduce a form of circular reasoning. Consequently, future research should assess constraints that are gathered independently, for example from process documentation, regulatory texts, or interviews with stakeholders.

Future work also aims at the generation of additional perspectives on, e.g., data, resource, and time, under constraints that involve these perspectives.

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A Full List of Constraints

Consulta

Approach	Metric	5 const	25 const	50 const
SC 3-FSA	Entropy	1.12 ±0.01	1.09 ±0.00	1.09 ±0.00
	2-gram	0.40 ±0.02	0.41 ±0.01	0.40 ±0.01
SC ∞-FSA	Entropy	1.10 ±0.01	1.05 ±0.02	1.05 ±0.03
	2-gram	0.40 ±0.02	0.40 ±0.00	0.40 ±0.00
S 3-FSA	Entropy	1.03 ±0.00	0.96 ±0.00	0.95 ±0.00
	2-gram	0.41 ±0.00	0.41 ±0.01	0.40 ±0.00
S ∞-FSA	Entropy	0.92 ±0.00	0.92 ±0.00	0.92 ±0.00
	2-gram	0.41 ±0.00	0.40 ±0.00	0.41 ±0.00
CVAE	Entropy	1.05 ±0.00	1.02 ±0.00	0.99 ±0.03
	2-gram	0.61 ±0.00	0.59 ±0.00	0.71 ±0.00
RuM	Entropy	0.57±0.00	0.57±0.00	0.57±0.00
	2-gram	0.86±0.00	0.87±0.00	0.88±0.00
$\mathcal{L}^{\mathcal{C}}_{test}$	Entropy	1.04	1.00	1.00

Production

Approach	Metric	5 const	25 const	50 const
SC 2-FSA	Entropy	1.32 ±0.01	1.27 ±0.02	1.27 ±0.00
	2-gram	0.38 ±0.00	0.42 ±0.00	0.43 ±0.00
SC ∞-FSA	Entropy	1.28 ±0.01	1.18 ±0.00	1.18 ±0.00
	2-gram	0.36 ±0.01	0.37 ±0.00	0.38 ±0.00
S 2-FSA	Entropy	1.27 ±0.00	1.23 ±0.00	1.23 ±0.00
	2-gram	0.42 ±0.00	0.42 ±0.00	0.42 ±0.00
S ∞-FSA	Entropy	1.22 ±0.00	1.02 ±0.00	1.02 ±0.00
	2-gram	0.32 ±0.00	0.37 ±0.01	0.35 ±0.00
CVAE	Entropy	1.15 ±0.00	1.26 ±0.00	0.98 ±0.00
	2-gram	0.39 ±0.00	0.49 ±0.00	0.48 ±0.00
RuM	Entropy	0.34±0.00	0.34±0.00	0.34±0.00
	2-gram	0.81±0.00	0.82±0.00	0.82±0.00
$\mathcal{L}^{\mathcal{C}}_{test}$	Entropy	1.15	1.14	1.14

Purchasing

Approach	Metric	5 const	25 const	50 const
SC 3-FSA	Entropy	1.40 ±0.04	1.40 ±0.03	1.37 ±0.03
	2-gram	0.45 ±0.00	0.40 ±0.00	0.40 ±0.00
SC ∞-FSA	Entropy	1.35 ±0.00	1.34 ±0.03	1.31 ±0.00
	2-gram	0.44 ±0.02	0.40±0.00	0.40 ±0.03
S 3-FSA	Entropy	1.35 ±0.02	1.37 ±0.01	1.37 ±0.00
	2-gram	0.44 ±0.01	0.40 ±0.00	0.40 ±0.00
S ∞-FSA	Entropy	1.22 ±0.06	1.06 ±0.00	1.04 ±0.00
	2-gram	0.44 ±0.00	0.40 ±0.00	0.40 ±0.00
CVAE	Entropy	1.28 ±0.00	1.12 ±0.00	1.06 ±0.00
	2-gram	0.61 ±0.00	0.61 ±0.00	0.64 ±0.00
RuM	Entropy	0.46±0.00	0.42±0.00	0.41±0.00
	2-gram	0.87±0.00	0.88±0.00	0.88±0.00
$\mathcal{L}^{\mathcal{C}}_{test}$	Entropy	1.30	1.20	1.18

Emergency Dept

Approach	Metric	5 const	25 const	50 const
SC 3-FSA	Entropy	1.42 ±0.01	1.34 ±0.00	1.30 ±0.00
	2-gram	0.30 ±0.00	0.32 ±0.02	0.34 ±0.02
SC ∞-FSA	Entropy	1.29 ±0.00	1.24 ±0.00	1.22 ±0.00
	2-gram	0.33 ±0.00	0.30 ±0.00	0.32 ±0.00
S 3-FSA	Entropy	1.18 ±0.00	1.13 ±0.00	1.18 ±0.00
	2-gram	0.21 ±0.00	0.32 ±0.01	0.28 ±0.02
S ∞-FSA	Entropy	1.06 ±0.00	1.05 ±0.00	1.07 ±0.00
	2-gram	0.20 ±0.00	0.20 ±0.00	0.18 ±0.02
CVAE	Entropy	1.16±0.02	1.12±0.02	1.10±0.02
	2-gram	0.32±0.05	0.39±0.01	0.42±0.01
RuM	Entropy	0.69±0.00	0.67±0.00	0.67±0.00
	2-gram	0.99±0.00	0.99±0.00	0.99±0.00
$\mathcal{L}^{\mathcal{C}}_{test}$	Entropy	1.30	1.34	1.34

Lending

Approach	Metric	5 const	25 const	50 const
SC 2-FSA	Entropy	0.92 ±0.01	0.93 ±0.00	0.92 ±0.00
	2-gram	0.04 ±0.00	0.19 ±0.00	0.19 ±0.00
SC ∞-FSA	Entropy	0.91 ±0.00	0.77 ±0.00	0.77 ±0.00
	2-gram	0.06 ±0.01	0.49 ±0.00	0.50 ±0.00
S 2-FSA	Entropy	0.91 ±0.00	0.81 ±0.00	0.82 ±0.02
	2-gram	0.06 ±0.01	0.24 ±0.01	0.24 ±0.02
S ∞-FSA	Entropy	0.85 ±0.00	0.73 ±0.00	0.73 ±0.00
	2-gram	0.04 ±0.00	0.44 ±0.00	0.44 ±0.00
CVAE	Entropy	0.81±0.01	0.70±0.01	0.70±0.01
	2-gram	0.12±0.02	0.34±0.02	0.35±0.04
RuM	Entropy	0.70±0.00	0.70±0.00	0.67±0.00
	2-gram	0.97±0.00	0.99±0.00	0.99±0.00
$\mathcal{L}^{\mathcal{C}}_{test}$	Entropy	0.93	0.82	0.81

Fig. 7: Trace entropy and 2-gram distance metrics across all five use cases for our framework with k infinite or finite (denoted as SC 2-, SC 3- and SC ∞-FSA) and for the baselines listed at page 10, namely the use of non-stochastic k-FSA (denoted as S 2-, S 3-, S ∞-FSA), CVAE and the RuM toolkit. Entries marked with "×" denote use cases where no compliant trace exists in the log, preventing every frameworks from being evaluable, with exception of SC-k FSA (cf. Table **??**). The rows for $\mathcal{L}^{\mathcal{C}}_{test}$ report the entropy for the test log.